

Probabilistic Forecasting Evaluation with Large Language Models

Aims:

The aim of this project is to systematically evaluate the ability of various large language models (e.g., GPT, Llama, Claude) to generate probabilistic and distributional forecasts on benchmark datasets. The evaluation will cover point predictions, full probability distributions, and associated uncertainty quantification, assessing both accuracy and calibration using proper scoring rules.

Background:

Large language models (LLMs) have revolutionised the way many intellectual tasks are being performed. Given their ability in storing and reorganising information, it is not impossible to believe that they may also represent very good solutions for general purpose distributional forecasting. However, rigorous evaluations of their performance are limited, if not non-existent. This project addresses the gap by implementing evaluation pipelines that benchmark a variety of LLMs against classical and contemporary probabilistic forecasting models. Attention will be given to calibration, sharpness, and efficiency of forecasted distributions using modern scoring metrics.

Early Deliverables

1. Survey recent literature on LLMs in probabilistic forecasting (e.g., probabilistic numeracy, uncertainty quantification, distributional prediction tasks)
2. Select representative LLMs to benchmark, including open-source and commercial APIs.
3. Identify simple simulative time-series benchmarks.
4. Identify benchmark datasets suitable for probabilistic evaluation (e.g., weather data, time series competitions, tabular uncertainty datasets).
5. Specify proper scoring rules for evaluation, such as the Continuous Ranked Probability Score (CRPS), Log-Score, and calibration metrics (PIT histograms, reliability diagrams).

6. Implement proof-of-concept code to query LLMs for probabilistic forecasts in various formats (e.g., quantiles, predictive intervals, full PDFs).
7. Preprocess datasets and design data wrappers to facilitate model predictions and structured forecast evaluation.
8. Provide an initial report reviewing LLMs' forecasting output formats and semantic interpretations.

Final Deliverables

1. Develop a full benchmarking framework to conduct large-scale comparison of LLMs in probabilistic/distributional forecasting.
2. Quantitatively evaluate each model using chosen scoring rules, and compare against dedicated probabilistic ML methods (e.g., quantile regression, deep ensembles).
3. Analyze calibration and efficiency of LLM forecasts across datasets and scenarios.
4. Produce a comprehensive report including results, interpretability analysis, and methodological insights.
5. Release all benchmarking code and protocols, aiming for reproducibility.
6. Optional: Prepare an R or Python package to wrap LLMs for probabilistic prediction tasks.
7. Optional: Submit findings for journal publication or conference presentation.

Suggested Extensions

- Explore post-hoc calibration strategies (e.g., conformal prediction, Bayesian postprocessing) for improving LLM output validity.
- Investigate computational costs and practical challenges in deploying probabilistic LLMs in real-world forecasting environments.

Reading

- Jordan, A., Krüger, F., " Lerch, S. (2019). Evaluating Probabilistic Forecasts with scoringRules. Journal of Statistical Software, 90(12).
- Gneiting, T., " Katzfuss, M. (2014). Probabilistic forecasting. Annual Review of Statistics and Its Application, 1, 125-151.

- Allen, S., et al. (2025). In-sample calibration yields conformal calibration guarantees. arXiv:2503.03841.

Prerequisites:

Strong Python skills, including APIs for LLMs (e.g., OpenAI, Huggingface). Interest in probabilistic modeling or uncertainty quantification is recommended.